

24.1

Area (1)

24.1.1 Area: NIELIT 2023 Feb Scientist C | Section D | Question: 111



A chord AB of a circle of radius 5.25 cm makes an angle 60° at the centre of the circle. Find the area of major segment.

- A. 168 cm^2 B. 42 cm^2
C. 84 cm^2 D. 78 cm^2

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24.2

Geometry (5)

24.2.1 Geometry: NIELIT 2021 Dec Scientist B - Section A: 26



In a parallelogram $ABCD$, AP and BP are the angle bisectors of $\angle DAB$ and $\angle ABC$. Find $\angle APB$:

- A. 85° B. 90° C. 94° D. 86°

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Answer key

24.2.2 Geometry: NIELIT 2023 Feb Scientist C | Section C | Question: 72



The size of a rhombus is increased by itself, How many times will its perimeter increase?

- A. Three times B. Four times
C. Two times D. None of the options

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24.2.3 Geometry: NIELIT 2023 Feb Scientist D | Section D | Question: 108



In a triangle ABC , $2ac \sin \left[\frac{1}{2}(A-B+C) \right]$ is equal to :

- A. $a^2 + b^2 - c^2$ B. $c^2 + a^2 - b^2$ C. $b^2 - c^2 - a^2$ D. $c^2 - a^2 - b^2$

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24.2.4 Geometry: NIELIT 2023 Feb Scientist D | Section D | Question: 109



The vertices of a triangle ABC are $A(4, 6)$, $B(1, 5)$ and $C(7, 2)$. A line is drawn to intersect sides AB and AC at D and E respectively such that

$AD : AB = AE : AC = 4$. Calculate the ratio of area of triangle ADE and area of triangle ABC .

- A. 1 : 16 B. 16 : 1 C. 15 : 16 D. 1 : 1

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24.2.5 Geometry: NIELIT 2023 Feb Scientist D | Section D | Question: 110



In a triangle XYZ , P and Q are points on XY , XZ respectively such that $XP = 2PY$, $XQ = 2QZ$, then the ratio, of area of $\triangle XPQ$ and area of $\triangle XYZ$ is :

- A. 4 : 9 B. 2 : 3 C. 3 : 2 D. 9 : 4

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Answer key 

Answer Keys

24.1.1	C	24.2.1	B	24.2.2	C	24.2.3	B	24.2.4	A
24.2.5	A								